Akoizo Original Practice - Form 5 - Answer key

Division B | Roller Coaster | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Target Time 40 s. The ball crosses at 43.2 s. What is the Time Score?

A. 185

B. 200

C. 215

D. 0

Correct: A

2. (2 points) Why is the ideal release height 2.7 r rather than 2.5 r for a rolling ball?

A. Some energy must go into the ball’s spin

B. Gravity is stronger in loops

C. The loop is enclosed

D. Friction is zero

Correct: A

3. (2 points) What is the first tiebreaker?

Unit: Rules and the four-part score.
Worked-case data: Target Time 45 s. A 52 x 26 x 72 cm device runs 44.6 s with both elevators and an 18.4 cm loop. Find the Run Score.
Separate investigation observations:
Design A: 15 cm loop, no elevators, size 30, time 215.
Design B: 20 cm loop, two elevators, size 15, time 210.
Target Time: 45 s for both.

A. Highest Loop Score

B. Highest Size Score

C. Longest run time

D. Fewest elevators

Correct: A

4. (2 points) Why must covered sections be clear?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

A. To add Size points

B. To speed the ball

C. The ball must remain visible at all times

D. To reduce friction

Correct: C

5. (4 points) Explain why elevators use IMA less than 1.

answer: The counterweight moves a short distance with a large force to lift the light ball a longer distance. (4 points).

6. (4 points) Calculate the acceleration of a solid ball on a 5° incline, ignoring friction.

answer: a = (5/7) x 9.8 x sin 5° about 0.61 m/s^2. (4 points).

7. (4 points) How should the pencil be removed at the start?

Unit: Testing and competition day.
Worked-case data: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?
Separate investigation observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

answer: Pull it from the track on “Go” without exerting any force on the ball. (4 points).

8. (4 points) What causes balls to stop on slow sections?

Unit: Hitting the Target Time.
Worked-case data: Calibration: base path 34 s; each slope-screw turn adds 0.8 s; removable spiral adds 9 s. Plan for a Target Time of 46 s.
Separate investigation observations:
Runs at setting A: 41.8, 42.1, 41.9 s.
Runs at setting B: 44.6, 45.4, 44.8 s.
Target Time: 45 s.

answer: Slopes too gentle for the friction, flat spots, sticky joints, or dips where the ball settles. (4 points).

9. (2 points) Why did 24 cm fail?

Supplied synthetic observations:
Release 24 cm: Ball fell from top of loop in 6 of 10 trials.
Release 28 cm: Ball completed loop 10 of 10.
Loop path radius: 9 cm.

A. It was below the ideal 24.3 cm and friction lowered speed further

B. Too much speed

C. The loop was enclosed

Correct: A

10. (6 points) Solve this case and explain your reasoning: Target Time 45 s. A 52 x 26 x 72 cm device runs 44.6 s with both elevators and an 18.4 cm loop. Find the Run Score.

reasoning: Size: (60 - 52) + (30 - 26) + (80 - 72) = 8 + 4 + 8 = 20.
Time: 44 full seconds x 5 = 220. Elevators: 2 x 50 = 100.
Loop: 18 whole cm x 3 = 54. Total = 20 + 220 + 100 + 54 = 394. (3 points).

conclusion: Run Score = 394 points. (3 points).

11. (6 points) Solve this case and explain your reasoning: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.

reasoning: Option 1: Size 5 + 5 + 5 = 15; Loop 22 x 3 = 66; total 81.
Option 2: Size 15 + 10 + 15 = 40; Loop 15 x 3 = 45; total 85.
Option 2 scores 4 more, if both hit the Target Time equally. (3 points).

conclusion: Option 2 edges ahead; both are close, so reliability decides. (3 points).

12. (2 points) Does the elevator earn points?

Supplied synthetic observations:
Elevator test: Counterweight drops 5 cm; ball rises 9 cm.
Requirement: Ball must rise at least 10 cm.
Team claim: Elevator earns 50 points.

A. No; the ball rises less than 10 cm

B. Yes; IMA is below 1

C. Yes; the counterweight moved

Correct: A

13. (2 points) Define “Gravitational potential energy” in the context of Roller Coaster.

answer: mgh, stored by height. (2 points).

14. (2 points) Define “Practice run” in the context of Roller Coaster.

answer: Unscored run during the 8 minutes. (2 points).

15. (2 points) Define “Switchback” in the context of Roller Coaster.

answer: Zigzag track that lengthens the path. (2 points).

16. (2 points) Define “Release height rule” in the context of Roller Coaster.

answer: h >= 2.7 r for a rolling solid sphere, no friction. (2 points).

17. (2 points) Define “Elevator Score” in the context of Roller Coaster.

answer: 50 points for each elevator type used successfully. (2 points).

18. (2 points) Define “Finish Line” in the context of Roller Coaster.

answer: Labeled line ending timing when crossed. (2 points).